Backward-Compatible Representation Learning with DualPrompt: Continual Learning and Feature Compatibility Analysis

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1 Abstract

Continual Learning aims to enable machine learning models to incrementally acquire new knowledge over time without catastrophically forgetting previously learned information. While many approaches focus on preserving task performance across sequential learning stages, less attention has been devoted to the backward compatibility of learned feature representations, a property that is crucial in real-world scenarios where previously extracted features cannot be recomputed.
The backward compatibility of representations learned through DualPrompt, a prompt-based continual learning method, is analyzed under a query–gallery evaluation protocol. Feature representations extracted by earlier model versions are stored in a gallery, while updated models are used to extract features for incoming queries. Compatibility between old and new representations is evaluated without re-encoding the gallery data.
To further investigate representation stability, a distillation-based loss is introduced to encourage alignment between features produced at consecutive tasks, promoting backward compatibility across model updates. Compatibility is evaluated both in-distribution and in an out-of-distribution setting, where the BloodMNIST dataset is used to construct query and gallery samples.
In addition to retrieval performance, the evolution of the feature space is analyzed through visualizations and Centered Kernel Alignment (CKA), providing insight into how representations change across tasks. The experimental results indicate that DualPrompt maintains a reasonable level of backward compatibility across learning stages, while feature alignment through distillation further stabilizes the representation space.

2 Introduction

Continual Learning (CL) addresses the problem of training machine learning models on a sequence of tasks while preserving knowledge acquired from previous learning stages. A major challenge in this setting is catastrophic forgetting, whereby performance on earlier tasks degrades significantly as new tasks are learned. Over the years, several approaches have been proposed to mitigate this issue, including replay-based methods, regularization strategies, and architectural modifications.

Beyond task performance retention, an increasingly relevant but less explored aspect of Continual Learning concerns the backward compatibility of feature representations. In many real-world applications, feature embeddings extracted by a model are stored and reused over time, for example in large-scale retrieval systems or biometric databases. In such scenarios, updating the model and re-extracting all stored features may be computationally infeasible or practically impossible. As a consequence, newly learned representations should remain compatible with those produced by earlier model versions.

Prompt-based methods for Continual Learning have recently gained attention due to their effectiveness and flexibility, particularly in combination with large pretrained models. Among these approaches, DualPrompt [1] introduces a mechanism that separates task-invariant and task-specific knowledge through the use of global and expert prompts. While DualPrompt has been primarily evaluated in terms of its ability to alleviate catastrophic forgetting, its impact on the backward compatibility of learned representations remains largely unexplored.

This project investigates the backward compatibility properties of feature representations learned using DualPrompt. The evaluation is conducted under a realistic query–gallery protocol, where feature embeddings extracted by previous model checkpoints are stored in a fixed gallery, and updated models are used to encode incoming queries. Compatibility is assessed without re-encoding the gallery data, following the setting proposed in recent studies on backward-compatible representation learning [2].

To further encourage representation stability across learning stages, an additional distillation-based loss is introduced to align feature embeddings produced at consecutive tasks, inspired by the knowledge distillation principle adopted in Learning without Forgetting [3]. This mechanism aims to improve backward compatibility by reducing the drift of the learned representation space during sequential training.

In addition to compatibility evaluation, the evolution of the feature space across tasks is analyzed through dimensionality reduction and representation similarity measures. In particular, Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE) are used to visualize the structure of the learned embeddings, both in two-dimensional and three-dimensional projections. These visualizations provide qualitative insight into how the representation space evolves as new tasks are learned. Fur-

thermore, Centered Kernel Alignment (CKA) is employed to quantify similarities between representations obtained at different stages of training. The robustness of the learned representations is also investigated under an out-of-distribution setting using samples from the BloodMNIST dataset within the same query–gallery evaluation protocol.

The remainder of this report is organized as follows. Section 3 reviews relevant literature on Continual Learning, prompt-based methods, and backward-compatible representations. Section 4 describes the DualPrompt approach and the adopted training procedure. The experimental setup and evaluation protocol are presented in Section 5. Results and discussion are provided in Sections 6, followed by concluding remarks in Section 7.

3 Related Work

Continual Learning has been extensively studied as a framework for enabling models to learn from a stream of tasks while mitigating catastrophic forgetting. Classical approaches include rehearsal-based methods, which retain a subset of past data, regularization-based techniques that constrain parameter updates, and architectural strategies that allocate separate model components to different tasks. While effective, these methods often introduce additional memory or computational overhead.

More recently, prompt-based approaches have emerged as a flexible alternative for Continual Learning, particularly in conjunction with large pre-trained models. Instead of modifying the backbone parameters, these methods learn task-specific prompts that guide the model's behavior. DualPrompt extends this idea by introducing both global prompts, which capture task-invariant knowledge, and expert prompts, which encode task-specific information. This separation has been shown to improve performance and reduce forgetting across sequential tasks.

In parallel, the concept of backward-compatible representation learning has gained attention in the context of realistic deployment scenarios. In applications such as image retrieval and face recognition, feature embeddings extracted by previous model versions are often stored and reused over long periods of time. Updating the feature extractor without reprocessing the entire database can lead to incompatibilities between old and new representations, resulting in performance degradation. Recent works have proposed evaluation protocols and learning strategies aimed at preserving compatibility across model updates.

Despite the growing interest in both prompt-based Continual Learning and backward-compatible representations, the interaction between these two research directions remains underexplored. In particular, existing evaluations of DualPrompt primarily focus on task accuracy and forgetting, without explicitly addressing the compatibility of learned feature embeddings across learning stages. This project aims to bridge this gap by analyzing DualPrompt under a backward compatibility evaluation protocol and by examining the evolution of the learned representation space during continual training.

4 Methodology

This section describes the learning framework adopted in this project. First, the DualPrompt approach is presented, including the prompt insertion mechanism and the associated optimization objective. Subsequently, the notion of backward-compatible representation learning and the adopted evaluation criteria are introduced.

4.1 DualPrompt

DualPrompt is a prompt-based Continual Learning method designed to adapt large pre-trained models to sequential tasks while mitigating catastrophic forgetting [1]. Instead of fine-tuning the backbone parameters, learnable prompts are introduced and optimized while keeping the pre-trained model frozen.

Given an input sequence represented by hidden states $\mathbf{h} \in \mathbb{R}^{L \times D}$, Prefix Tuning (Pre-T) splits a prompt $\mathbf{p}$ into key and value components, $\mathbf{p}_K$ and $\mathbf{p}_V$, where:

$$\mathbf{p}_K, \mathbf{p}_V \in \mathbb{R}^{\frac{L_p}{2} \times D}.$$

These components are prepended to the key and value sequences of the multi-head self-attention (MSA) module, while the query remains unchanged:

$$f_{\text{prompt}}^{\text{Pre-T}}(\mathbf{p}, \mathbf{h}) = \text{MSA}\left(\mathbf{h}_Q, [\mathbf{p}_K; \mathbf{h}_K], [\mathbf{p}_V; \mathbf{h}_V]\right).$$

This formulation preserves the original sequence length of the hidden representation.

DualPrompt introduces two types of prompts:

- *Global prompts* $\mathbf{v}_g$, shared across all tasks and intended to capture task-invariant knowledge.
- *Expert prompts* $\mathbf{v}_e^{(t)}$, which are specific to task t and encode task-dependent information.

In addition, each task is associated with a learnable task key $\mathbf{k}_t$, which is used to select the appropriate expert prompt at inference time. For an input sample $\mathbf{x} \in \mathcal{D}_t$, the optimization objective for task t is defined as:

$$\min_{\mathbf{v}_g, \mathbf{v}_e^{(t)}, \mathbf{k}_t, \phi} \mathcal{L}\big(f_\phi(f_{\mathbf{v}_g, \mathbf{v}_e^{(t)}}(\mathbf{x})), y\big) + \lambda\, \mathcal{L}_{\text{match}}(\mathbf{x}, \mathbf{k}_t), \quad \mathbf{x} \in \mathcal{D}_t, \tag{1}$$

where $\mathcal{L}$ denotes the cross-entropy loss, λ is a scalar balancing factor, and ϕ represents the parameters of the classification head.

The matching loss encourages the task key $\mathbf{k}_t$ to become closer to examples from the corresponding task than to those from other tasks. Following prior work, the matching loss is defined as:

$$\mathcal{L}_{\text{match}}(\mathbf{x}, \mathbf{k}_t) = \gamma\big(q(\mathbf{x}), \mathbf{k}_t\big), \quad \mathbf{x} \in \mathcal{D}_t, \tag{2}$$

where $q(\cdot)$ is a query function and $\gamma(\cdot, \cdot)$ denotes cosine similarity. The query function is defined as:

$$q(\mathbf{x}) = f(\mathbf{x})[0],$$

corresponding to the feature vector associated with the [CLS] token of the pre-trained model.

At inference time, for a test sample $\mathbf{x}$, the expert prompt is selected by identifying the task key that maximizes cosine similarity:

$$t^* = \arg\min_t \gamma\big(q(\mathbf{x}), \mathbf{k}_t\big).$$

The corresponding expert prompt $\mathbf{v}_e^{(t^*)}$ is then used to compute the final representation.

4.2 Backward-Compatible Representation Learning

Backward-compatible representation learning aims at ensuring that feature embeddings extracted by an updated model remain compatible with those produced by a previous model version [2]. In a strict sense, a new model ϕ_{new} is backward compatible with an old model ϕ_{old} if the following conditions hold:

$$\begin{aligned} d\big(\phi_{\text{new}}(\mathbf{x}_i), \phi_{\text{old}}(\mathbf{x}_j)\big) &\geq d\big(\phi_{\text{old}}(\mathbf{x}_i), \phi_{\text{old}}(\mathbf{x}_j)\big), \\ &\qquad \forall (i,j) \text{ s.t. } y_i \neq y_j, \\ d\big(\phi_{\text{new}}(\mathbf{x}_i), \phi_{\text{old}}(\mathbf{x}_j)\big) &\leq d\big(\phi_{\text{old}}(\mathbf{x}_i), \phi_{\text{old}}(\mathbf{x}_j)\big), \\ &\qquad \forall (i,j) \text{ s.t. } y_i = y_j. \end{aligned} \tag{3}$$

where $d(\cdot,\cdot)$ denotes a distance function in the embedding space.

These constraints formalize the requirement that the updated model should not reduce inter-class separability nor increase intra-class distances when compared against representations generated by the old model. The trivial solution $\phi_{\text{new}} = \phi_{\text{old}}$ satisfies this condition but is typically excluded in practice due to architectural or training differences between model versions.

Since the strict criterion in Eq. (3) is computationally intractable at scale, an empirical backward compatibility criterion is commonly adopted.
Let $M(\phi_q, \phi_g; Q, D)$ denote an evaluation metric computed on a query set Q and a gallery set D, where ϕ_q and ϕ_g are used to extract features for queries and gallery samples, respectively. Backward compatibility is empirically achieved if:

$$M(\phi_{\text{new}}, \phi_{\text{old}}; Q, D) > M(\phi_{\text{old}}, \phi_{\text{old}}; Q, D). \tag{4}$$

This criterion reflects a realistic deployment scenario in which gallery features extracted by an old model are kept fixed, and only the query encoder is updated. If performance improves under this constraint, the model update can be considered backward compatible and backfill-free.

5 Experimental Setup

This section describes the experimental configuration adopted to evaluate DualPrompt in a class-incremental setting and to analyze backward compatibility of learned representations.

5.1 Dataset and Continual Learning Protocol

Experiments are conducted on CIFAR-100, which contains 100 classes of natural images. The dataset is divided into 10 sequential tasks following a 10×10 class-incremental split, where each task introduces 10 new classes. The resulting sequence is denoted as $\{\mathcal{D}_1, \ldots, \mathcal{D}_{10}\}$.

Training follows the standard class-incremental learning protocol: at step t, only data from the current task $\mathcal{D}_t$ is available, and no samples from previous tasks are stored or replayed. After each task, the model is evaluated on all classes observed up to that stage.

5.2 Model Configuration

A Vision Transformer ViT-B/16 pre-trained on ImageNet is adopted as the backbone model. The backbone parameters remain frozen throughout training.

DualPrompt is implemented using global prompts, task-specific expert prompts, and task keys, following the Prefix Tuning formulation described in Section 4.1. Only the prompt parameters, task keys, and the classification head are optimized during continual learning.

The model is trained sequentially across tasks using cross-entropy loss combined with the matching loss defined in Eq. (1). Performance is measured in terms of classification accuracy.

5.3 Backward Compatibility Evaluation Protocol

Backward compatibility is evaluated under a cross-dataset query–gallery protocol inspired by realistic deployment scenarios. While DualPrompt is trained sequentially on CIFAR-100 (10×10 split), backward compatibility is verified using the CIFAR-10 test set as a fixed evaluation benchmark.

After training on task t, the model $\phi^{(t)}$ is used to extract feature embeddings from the CIFAR-10 test set. These embeddings are stored to form the gallery representations corresponding to that training stage.

For a pair of training stages (t_q, t_g) with $t_q \geq t_g$, the evaluation proceeds as follows:

- **Gallery construction:** CIFAR-10 test samples are encoded using model $\phi^{(t_g)}$. Token-level features are averaged and ℓ_2-normalized:

$$\mathbf{z} = \frac{\text{mean}(\mathbf{H}, \text{dim} = 1)}{\|\text{mean}(\mathbf{H}, \text{dim} = 1)\|_2}.$$

- **Query encoding:** the same CIFAR-10 test samples are encoded using model $\phi^{(t_q)}$ and normalized in the same manner.

- **Similarity computation:** cosine similarity is computed between query and gallery embeddings:

$$S = \mathbf{Z}_q \mathbf{Z}_g^\top.$$

- **k-NN classification:** for each query embedding, the top-K most similar gallery samples are selected. The predicted label is obtained by majority voting among the K nearest neighbors.

Classification accuracy is computed as:

$$\mathrm{Acc}(t_q, t_g) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\big(\hat{y}_i = y_i\big),$$

where N denotes the number of CIFAR-10 test samples.

The resulting compatibility scores are stored in a lower-triangular compatibility matrix $\mathbf{C} \in \mathbb{R}^{T \times T}$:

$$\mathbf{C}_{t_q, t_g} = \mathrm{Acc}(t_q, t_g), \quad t_q \geq t_g.$$

Diagonal entries correspond to standard evaluation using the same model for both query and gallery, while off-diagonal entries quantify backward compatibility across different training stages. Since the gallery embeddings are not recomputed when evaluating $\phi^{(t_q)}$ against earlier stages, this protocol directly reflects the empirical backward compatibility criterion defined in Eq. (4).

5.4 Feature Alignment via Distillation

In addition to the standard DualPrompt training procedure, an additional experiment introduces a feature-level distillation term aimed at improving backward compatibility of the learned representations.

The motivation is to reduce the drift of the embedding space across sequential tasks. Inspired by the knowledge distillation principle adopted in Learning without Forgetting [3], the feature embeddings produced by the current model are encouraged to remain aligned with those generated by the model obtained at the previous task.

Let f_{new} denote the feature embedding produced by the current model after training on task t, and f_{old} the embedding produced by the frozen model obtained after task $t-1$. The training loss combines the standard classification objective with a feature alignment term:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \, \mathcal{L}_{dist}, \tag{5}$$

where $\mathcal{L}_{CE}$ denotes the cross-entropy classification loss and λ controls the relative strength of the distillation term.

The distillation loss is defined using cosine similarity between feature embeddings:

$$\mathcal{L}_{dist} = 1 - \frac{f_{\text{new}} \cdot f_{\text{old}}}{\|f_{\text{new}}\| \, \|f_{\text{old}}\|}. \tag{6}$$

This formulation penalizes deviations between the feature embeddings produced by consecutive model versions. As a result, the current model is encouraged to preserve the geometric structure of the previously learned representation space while adapting to new tasks.

The effectiveness of this feature alignment strategy is evaluated using the same query–gallery backward compatibility protocol described in Section 5, enabling direct comparison with the baseline DualPrompt training configuration. task.

5.5 Feature Space Analysis Protocol

To analyze the evolution of the learned representation space across tasks, both qualitative and quantitative analyses are performed.

Dimensionality reduction techniques are used to visualize the structure of the feature embeddings. In particular, Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE) are used to obtain two-dimensional and three-dimensional projections of the embedding space.

While PCA provides a linear projection preserving global variance, t-SNE emphasizes local neighborhood structure and cluster compactness.

In addition to visual inspection, representation similarity across tasks is quantified using Centered Kernel Alignment (CKA), which measures the similarity between feature spaces produced by different model checkpoints.

5.6 Out-of-Distribution Evaluation

To evaluate the robustness of the learned representations under distribution shift, an additional compatibility experiment is performed using the BloodMNIST dataset.

While the model is trained on CIFAR-100 following the continual learning protocol, feature embeddings are extracted from BloodMNIST samples to construct both query and gallery sets.

The same cosine-similarity-based k-NN evaluation protocol used in the in-distribution setting is applied to measure compatibility between representations produced at different training stages.

This experiment allows assessing whether the representation space learned through continual learning remains coherent when evaluated on data outside the training distribution.

5.7 Evaluation with 5-Task, 20-Class Splits

To investigate the effect of task granularity on backward compatibility, an additional experiment is conducted by reorganizing CIFAR-100 into 5 sequential tasks, each containing 20 classes instead of the standard 10-class-per-task setup.

The model is trained following the same continual learning protocol, and feature embeddings are extracted after each task to construct both query and gallery sets.

Compatibility between embeddings from different tasks is measured using the same cosine-similarity-based k-NN evaluation procedure as in the standard 10-task setup.

This experiment allows assessing whether increasing the number of classes per task affects the stability and structural coherence of the learned feature space across sequential updates.

6 Results

This section presents the empirical evaluation of DualPrompt under multiple experimental settings. The analysis focuses on: (i) the quantitative assessment of backward compatibility through compatibility matrices, (ii) the qualitative inspection of the learned feature space via dimensionality reduction (t-SNE and PCA), (iii) the effect of feature-level distillation on representation stability, and (iv) the robustness of learned embeddings under an out-of-distribution (OOD) scenario.

The goal is to determine whether the learned representations remain stable and structurally consistent as new tasks are introduced, while preserving discriminative power.

6.1 Backward Compatibility Matrix

Backward compatibility across sequential tasks is summarized using a lower-triangular compatibility matrix:

$$\mathbf{C} = \begin{pmatrix} M(\phi_{Q_1}, \phi_{G_1}) & 0 & \cdots & 0 \\ M(\phi_{Q_2}, \phi_{G_1}) & M(\phi_{Q_2}, \phi_{G_2}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ M(\phi_{Q_T}, \phi_{G_1}) & M(\phi_{Q_T}, \phi_{G_2}) & \cdots & M(\phi_{Q_T}, \phi_{G_T}) \end{pmatrix} \tag{7}$$

where:

- ϕ_{Q_t} denotes the model used to extract query features after training on task t,
- ϕ_{G_k} denotes the model that generated the stored gallery features after task k,
- $M(\cdot, \cdot)$ represents the evaluation metric, defined as cosine-similarity-based k-NN accuracy ($K = 5$).

The generic element of the matrix is defined as:

$$\mathbf{C}_{t,k} = M(\phi_{Q_t}, \phi_{G_k}), \qquad \text{with } t \geq k. \tag{8}$$

The matrix is lower triangular because compatibility is evaluated only when the query model corresponds to a later training stage than the gallery model.

Interpretation. Diagonal elements ($t = k$) correspond to standard evaluation, where both query and gallery representations are extracted from the same model version. Off-diagonal elements ($t > k$) measure backward compatibility, assessing whether updated representations remain compatible with previously stored features. Values highlighted in red indicate cases where compatibility is not maintained.

6.2 Feature Space Visualization and Representation Similarity

The structure of the learned embedding space across sequential tasks is analyzed using both dimensionality reduction and representation similarity measures. Following the approach of liu2022cores, backward compatibility is assessed by examining the stationarity of feature distributions across tasks. In this framework, stable embeddings maintain similar distributional structure as new tasks are introduced. Dimensionality reduction plots are generated to visually inspect the evolution and alignment of feature embeddings over sequential updates.

Dimensionality Reduction. Feature embeddings are first ℓ_2-normalized and then projected into low-dimensional spaces using two complementary techniques:

- **t-Distributed Stochastic Neighbor Embedding (t-SNE)**: non-linear projection preserving local neighborhood structure. Embeddings are visualized in 2D to inspect cluster compactness, inter-class margins, and potential distortions introduced by aggressive dimensionality reduction.
- **Principal Component Analysis (PCA)**: linear projection into 2D, capturing global variance of the embeddings and complementing t-SNE visualizations by highlighting overall geometric structure.

Both t-SNE and PCA visualizations are generated independently at the end of each task. These projections allow a qualitative assessment of how new tasks affect the representation space of previously learned classes.

Centered Kernel Alignment (CKA). To quantitatively measure similarity between embeddings obtained at different tasks, CKA [4] is computed. Given two embedding matrices $X \in \mathbb{R}^{N \times d}$ and $Y \in \mathbb{R}^{N \times d}$ from tasks t and s, the linear CKA similarity is defined as:

$$\text{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \, \|Y^\top Y\|_F}, \tag{9}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. Values of CKA range from 0 to 1, with 1 indicating identical representation spaces. High CKA values across tasks indicate stability and alignment of learned features, while lower values reveal drift in the embedding space.

For visualization, the CKA scores are arranged in a heatmap similar to the backward compatibility matrix, allowing easy inspection of feature similarity across all task pairs.

Summary. The combination of t-SNE, PCA, and CKA provides a comprehensive picture of feature evolution:

- *t-SNE* highlights local neighborhood preservation and clustering quality.
- *PCA* reveals global structure and variance alignment across tasks.
- *CKA* quantitatively measures similarity between embedding spaces at different stages, complementing qualitative visualizations.

This framework is applied to both baseline DualPrompt and experiments with feature-level distillation, as well as to out-of-distribution evaluation using BloodMNIST, to assess representation robustness and backward compatibility.

6.3 Baseline DualPrompt Backward Compatibility

This section reports the backward compatibility evaluation for the baseline DualPrompt model, without any additional distillation or feature-alignment losses. Both quantitative and qualitative analyses are presented.

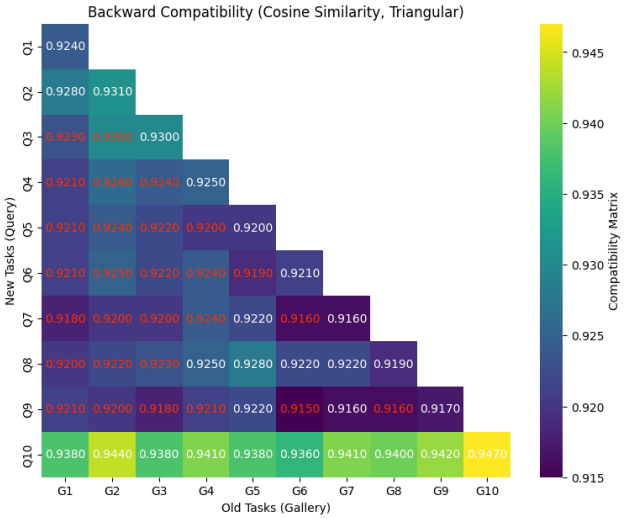

Figure 1: Backward Compatibility Matrix for the baseline DualPrompt. Rows correspond to updated models (Query), while columns correspond to stored feature encoders (Gallery).

Quantitative Analysis. Figure 1 shows high compatibility scores across tasks. Diagonal values remain stable, indicating negligible degradation in standard classification performance. Off-diagonal entries generally exceed 0.92, demonstrating that later embeddings preserve alignment with previously learned features. Minor fluctuations occur for certain task pairs, but these do not indicate systematic drift. Overall, the matrix confirms that the baseline DualPrompt maintains stable and interoperable feature representations across sequential tasks.

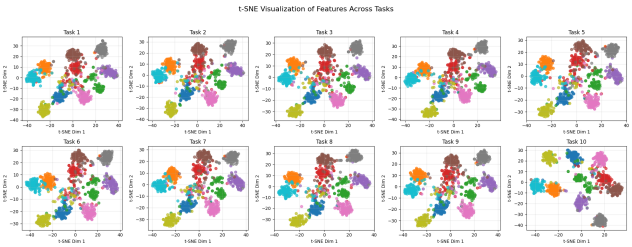

Figure 2: 2D t-SNE visualization of feature embeddings after each task for the baseline DualPrompt. Colors indicate class labels.

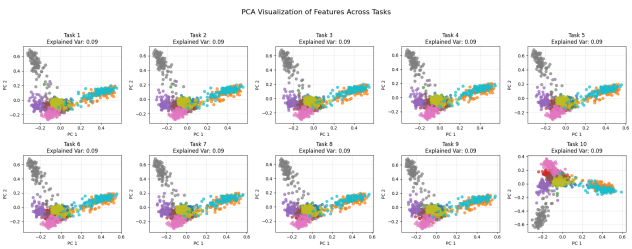

Figure 3: 2D PCA visualization of feature embeddings after each task for the baseline DualPrompt. Colors indicate class labels.

Qualitative Analysis. Figures ?? and 3 show that, from Task 1 to Task 9, the distribution of feature embeddings remains largely invariant. Minor variations are visible, reflecting slight feature drift, which is also captured quantitatively in the off-diagonal entries of the compatibility matrix.

For Task 10, however, both t-SNE and PCA visualizations suggest a global rototranslation of the embedding space compared to previous tasks: the relative positions of clusters are shifted and rotated, even though their internal structure and inter-class separability remain well-preserved. Interestingly, despite this apparent geometric transformation, the backward compatibility scores in the matrix are actually higher than in previous tasks.

This apparent contradiction arises because the compatibility metric is based on cosine similarity in the original high-dimensional embedding space, which is invariant to global rotations or translations of the entire feature set. As a result, a rigid transformation of the embedding space can occur without negatively affecting the similarity-based k-NN performance. This explains why the Task 10 embeddings appear rotated in low-dimensional projections but still achieve strong compatibility with earlier features.

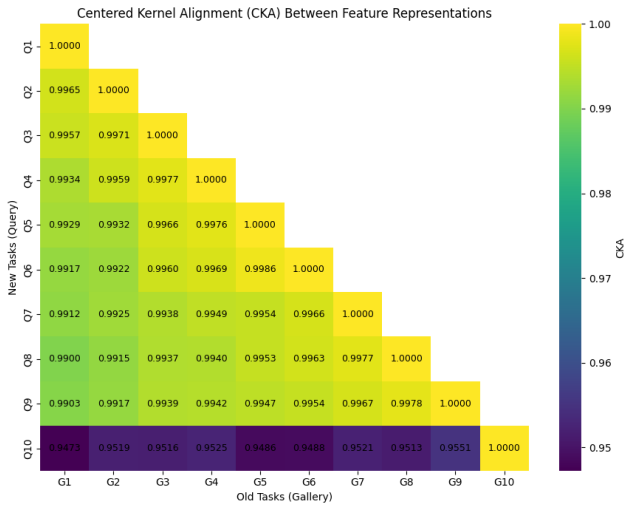

Figure 4: CKA similarity heatmap for baseline DualPrompt embeddings across tasks. Diagonal values are 1 by definition.

Representation Similarity Analysis (CKA). Figure 4 shows that embeddings remain highly aligned across tasks. Diagonal values are 1, as expected. Off-diagonal similarities between Tasks 1–9 are above 0.99, reflecting minimal drift. For Task 10, the similarity with earlier tasks slightly decreases to 0.95, consistent with the global rototranslation observed in t-SNE and PCA visualizations, but still indicating strong alignment of the embedding space.

Summary. The baseline DualPrompt achieves strong backward compatibility without explicit feature-alignment losses. Both quantitative metrics and visualizations indicate that the learned embeddings maintain discriminative power and structural coherence across incremental learning stages.

6.4 DualPrompt with Feature Distillation Backward Compatibility

This section reports the backward compatibility evaluation for the DualPrompt model trained with the feature-level distillation loss. Both quantitative and qualitative analyses are presented.

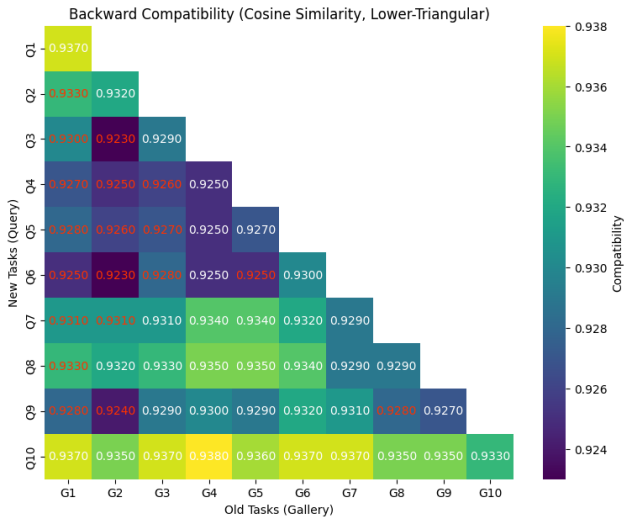

Figure 5: Backward Compatibility Matrix for DualPrompt with feature distillation. Rows correspond to updated models (Query), while columns correspond to stored feature encoders (Gallery).

Quantitative Analysis. Figure 5 shows consistently high compatibility scores across tasks. Diagonal values are stable, and off-diagonal entries generally remain very close to the corresponding diagonals. The percentage of compatible values (entries greater than or equal to the diagonal) is 57.78%, indicating stronger backward compatibility compared to the baseline. Overall, the distillation loss helps to further align the embeddings of newly learned tasks with previously stored features.

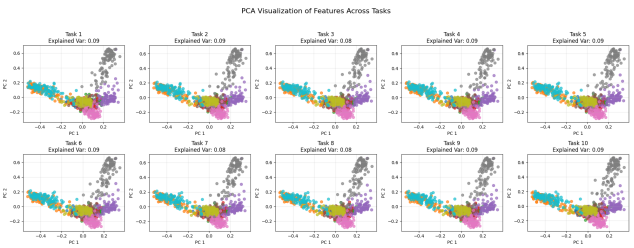

Figure 6: 2D PCA visualization of feature embeddings after each task for DualPrompt with feature distillation. Colors indicate class labels.

Qualitative Analysis. Figure 6 show that, from Task 1 to Task 10, the distribution of feature embeddings remains largely invariant. Minor variations are visible, reflecting slight feature drift. Unlike the baseline, even Task 10 embeddings remain fully aligned with the previous tasks, without any apparent rototranslation. Both t-SNE and PCA visualizations confirm that the embedding space preserves internal cluster structure and inter-class separability throughout incremental updates.

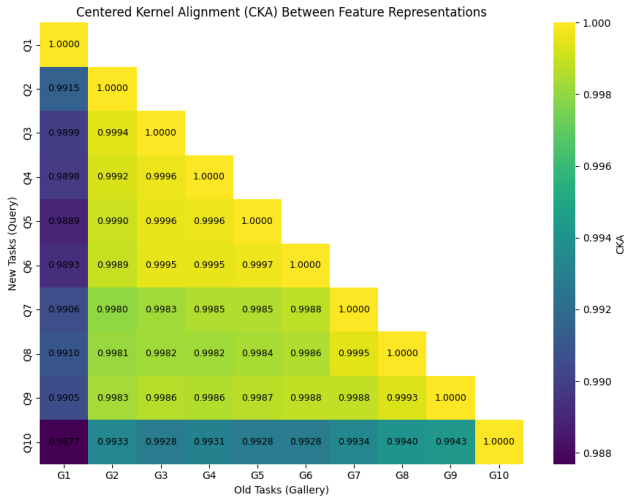

Figure 7: CKA similarity heatmap for DualPrompt with feature distillation embeddings across tasks. Diagonal values are 1 by definition.

Representation Similarity Analysis (CKA). Figure 7 confirms that embeddings remain highly aligned across tasks. All off-diagonal CKA values are around 0.99, indicating minimal drift and strong alignment of the embedding space, including Task 10.

Summary. Adding feature-level distillation improves backward compatibility compared to the baseline. Quantitative metrics, visualizations, and CKA analysis all indicate that the learned embeddings maintain discriminative power and structural coherence across incremental learning stages, with even the final task fully aligned with previous feature distributions.

6.5 DualPrompt Out-of-Distribution (OOD) Backward Compatibility

This section reports the backward compatibility evaluation for the DualPrompt model on out-of-distribution (OOD) tasks. Both quantitative and qualitative analyses are presented.

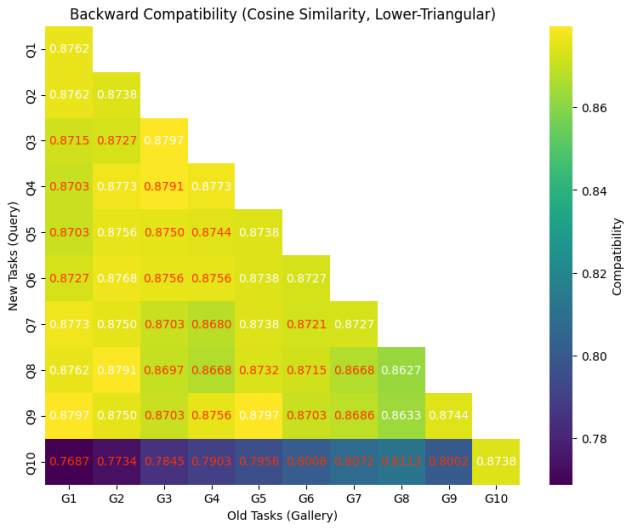

Figure 8: Backward Compatibility Matrix for DualPrompt on OOD tasks. Rows correspond to updated models (Query), while columns correspond to stored feature encoders (Gallery).

Quantitative Analysis. Figure 8 shows lower compatibility scores compared to the baseline and distillation experiments. Diagonal values remain relatively stable, but off-diagonal entries are on average 0.87, reflecting increased drift in embeddings when evaluating OOD tasks. The percentage of compatible values (entries greater than or equal to the diagonal) is 31%, lower than the 37% observed in the baseline.

Notably, Task 10 behaves differently compared to previous experiments: its backward compatibility scores with earlier tasks are the lowest, around 0.79–0.80, indicating that the final OOD task introduces the largest shift in the embedding space.

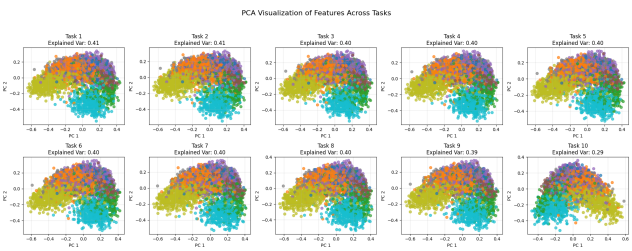

Figure 9: 2D PCA visualization of feature embeddings after each OOD task for DualPrompt. Colors indicate class labels.

Qualitative Analysis. Figure 9 shows that, for Tasks 1–9, the distributions of feature embeddings remain broadly similar, with minor variations reflecting moderate feature drift. For Task 10, however, the embedding distribution exhibits a clear rototranslation relative to the previous tasks, consistent with the drop in backward compatibility scores. The internal cluster structure and inter-class separability are preserved, but the global position of Task 10 embeddings differs noticeably.

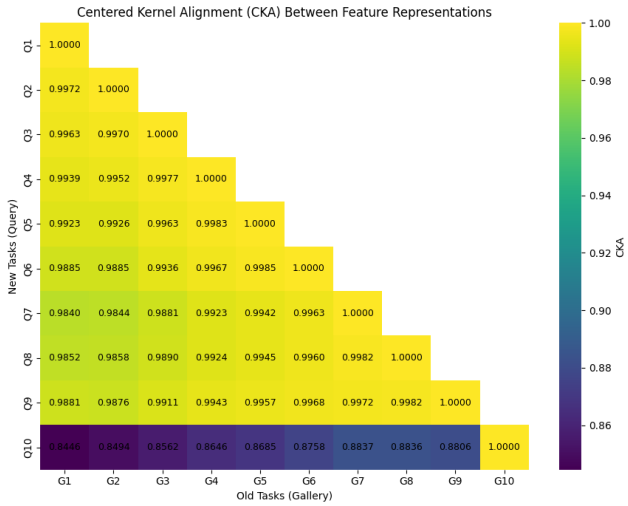

Figure 10: CKA similarity heatmap for DualPrompt embeddings across OOD tasks. Diagonal values are 1 by definition.

Representation Similarity Analysis (CKA). Figure 10 shows that embeddings for Tasks 1–9 remain highly aligned, with off-diagonal CKA values similar to those in the baseline (0.99). Task 10, however, exhibits reduced similarity with earlier tasks, reflecting the rototranslation observed in PCA and the lower compatibility scores, with CKA values around 0.80.

Summary. Backward compatibility is more challenging under OOD conditions. While early tasks maintain aligned embeddings, the final OOD task introduces significant shifts in the embedding space, leading to lower compatibility and CKA values. PCA visualizations confirm that Task 10 undergoes a global rototranslation, highlighting the difficulty of preserving feature alignment for out-of-distribution tasks.

6.6 DualPrompt Backward Compatibility with 5-Task, 20-Class Splits

This section reports the backward compatibility evaluation for the DualPrompt model when CIFAR-100 is divided into 5 sequential tasks of 20 classes each. Both quantitative and qualitative analyses are presented.

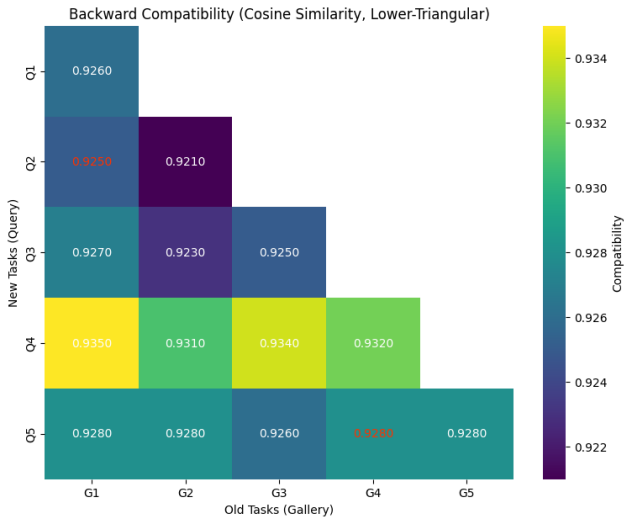

Figure 11: Backward Compatibility Matrix for DualPrompt with 5-task, 20-class splits. Rows correspond to updated models (Query), while columns correspond to stored feature encoders (Gallery).

Quantitative Analysis. Figure 11 shows that compatibility scores remain high across tasks, with most entries around 0.92–0.93. Only a few off-diagonal values are slightly lower: $M(\phi_{Q_2}, \phi_{G_1}) = 0.925$ (vs. 0.926) and $M(\phi_{Q_5}, \phi_{G_4}) =$ 0.928 (vs. 0.932). Overall, the matrix indicates that feature embeddings remain strongly aligned across task updates, confirming robust backward compatibility under this alternative task granularity.

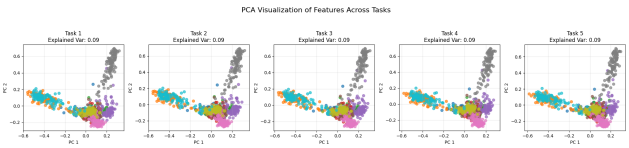

Figure 12: 2D PCA visualization of feature embeddings after each task for the 5-task, 20-class configuration. Colors indicate class labels.

Qualitative Analysis. Figure 12 shows that the distributions of feature embeddings remain largely invariant from Task 1 to Task 5. Minor variations are visible, reflecting slight feature drift, but internal cluster structure and inter-class separability are well preserved. Unlike in some other experiments, no global rototranslation is observed, and the embedding space remains aligned across all tasks.

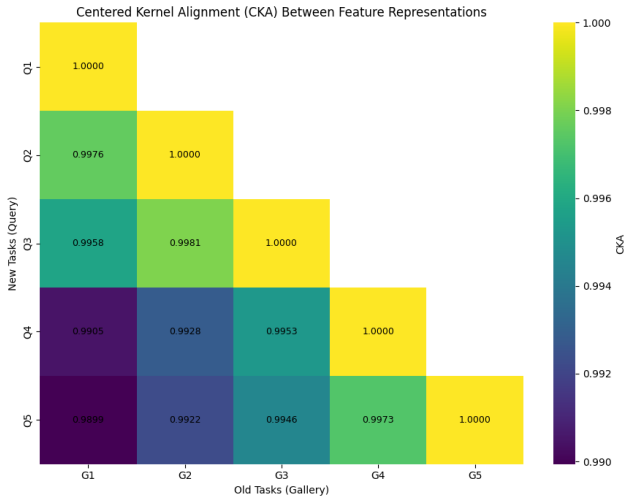

Figure 13: CKA similarity heatmap for DualPrompt embeddings across the 5-task, 20-class splits. Diagonal values are 1 by definition.

Representation Similarity Analysis (CKA). Figure 13 confirms that embeddings remain highly aligned across all tasks. Off-diagonal CKA values are consistently around 0.99, indicating minimal drift and strong preservation of the learned feature space.

Summary. The 5-task, 20-class split maintains strong backward compatibility. Both the compatibility matrix and PCA/CKA analyses indicate that embeddings remain well-aligned across sequential updates, demonstrating that increasing the number of classes per task does not adversely affect the structural coherence of the learned representation space.

7 Conclusions

In this work, we evaluated the backward compatibility of the DualPrompt model under multiple incremental learning settings, including standard 10-task splits, feature-level distillation, out-of-distribution scenarios, and coarser 5-task splits with more classes per task.

The main findings can be summarized as follows:

- **Baseline backward compatibility:** The standard DualPrompt model maintains high compatibility across tasks, with off-diagonal entries above 0.92 for most task pairs. Despite minor rototranslations observed in PCA and t-SNE for the final task, cosine-similarity-based k-NN performance remains strong, confirming that embeddings preserve discriminative structure even under geometric transformations.
- **Effect of feature-level distillation:** Incorporating a feature-level distillation loss further improves backward compatibility. Diagonal and off-diagonal compatibility scores remain consistently high, CKA values show minimal drift, and PCA visualizations indicate that embeddings across all tasks, including the last one, remain fully aligned without noticeable rotations or translations. This demonstrates that distillation effectively stabilizes the representation space during incremental learning.
- **Out-of-distribution evaluation:** When evaluated on BloodMNIST, compatibility scores decrease and the last task shows the largest shift in the embedding

space. PCA and CKA analyses confirm a global rototranslation for the final task, highlighting the challenges of preserving backward compatibility under distribution shifts. Nonetheless, early tasks remain well-aligned, indicating partial robustness of the learned features.

- **Impact of task granularity (5-task, 20-class splits):** Increasing the number of classes per task does not negatively affect backward compatibility. The compatibility matrix remains high ($\approx 0.92 - 0.93$), PCA embeddings show minimal drift, and CKA confirms strong alignment across all tasks. This suggests that DualPrompt can handle coarser task divisions while maintaining structural coherence in the learned feature space.

Overall, these results demonstrate that DualPrompt produces stable and discriminative embeddings across sequential updates. Feature-level distillation provides additional alignment benefits, especially for later tasks, while OOD evaluation highlights limitations under distribution shifts. Moreover, the model is robust to changes in task granularity, maintaining backward compatibility even when the number of classes per task is increased.

A possible explanation for the strong stability of the learned features is related to the behavior of attention in transformer backbones. As reported by [5], shallower layers exhibit a wider range of attention distances and focus on both localized and global regions, while deeper layers tend to encode more global contextual information. In DualPrompt, prompts are attached to multiple self-attention layers rather than only at the input sequence, which may enable the model to preserve rich intermediate representations across tasks. This mechanism could underlie the observed robustness and backward compatibility of embeddings, as it supports stable and discriminative features throughout the network.

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